THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/1 **CHEMISTRY 1**

TIME: 3:00 Hours **MARKING SCHEME** **August, 2024**

SECTION: A (16marks)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	C	E	A	B	D	D	E	A	B

*1mark**@=10 total marks***

2.

LIST A	**i**	**ii**	**iii**	**iv**	**v**	**vi**
LIST B	E	G	H	A	C	B

*1mark**@=06 total marks***

3. (a) Chemistry laboratory apparati made up by ceramic materials
   i. Mortar and pestle
   ii. Crucible and lid
   iii. Evaporating dish *1mark**@=04total marks***
   iv. White tile

(b) Soln.

Data given.

Quantity of electricity (Q) = 9650 Coulombs

Molar Mass of Ag = 108

G.M.V = $224dm^3$**(0.5 Mark)**

but

$Ag_{(ag)}^{+} + e^{-} \rightarrow Ag_{(s)}$ Liberated at Cathode **(0.5 Mark)**

1e = 1F

1F = 1mol of Ag = 108g

but

IF = 96500C

96500C = 108g of Ag

9650C = ?

9650C X 108g……..**(0.5 Mark)**

96500C

Mass of Ag(Silver) = 10.8g**…….01 mark**

$4OH^-_{(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)} + 4\ e^-$ (Liberated at Anode)**(0.5 Mark)**

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$4e^- = 4F$

$4F = 1mol$ of O_2 at s.t.p

$1mol$ of any gas at s.t.p $= 22.4dm^3$ **(0.5 Mark)**

$4F = 22.4dm^3$

$4(96500C) = 22.4dm^3$ of O_2 **(0.5 Mark)**

$9650C = ?$

$\frac{9650 \text{ coulombs} \times 22.4dm^3}{4(96500) \text{ coulombs}} = 0.56dm^3$

∴Volume of Oxygen Liberated at s.t.p $= 0.56dm^3$ **OR** $560cm^3$ **(01 Mark)**

4. (a)Decomposition of copper (II) sulphate

$CuSO_{4(s)} \xrightarrow[\Delta]{} CuO_{(s)} + SO_{3(g)}$

$Decomposition\ of\ copper\ (II) nitrate$

$2Cu(NO_3)_{2(s)} \xrightarrow[\Delta]{} 2CuO_{(s)} + 4NO_{2(g)} + O_{2(g)}$ *1mark* ***@=02 total marks***

(b) First stage: To calculate empirical formula of a compound

Data

Mass of carbon=0.6g

Mass of hydrogen=0.1g

RAM of carbon=12

RAM of Hydrogen=1

Required: Empirical formula

Symbol of an element	C	H
Mass of an element	0.6	0.1
RAM of an element	12	1
$\frac{Mass\ of\ an\ element}{RAM\ of\ an\ element}$	$\frac{0.6}{12} = 0.05$	$\frac{0.1}{1} = 0.1$
Divide by smallest value	$\frac{0.05}{0.05} = 1$	$\frac{0.1}{0.05} = 2$
Whole number	1	2

02marks

∴The empirical formula of a compound is CH_2**……..01mark**

Second stage: To calculate molecular mass of a compound

MM=2 X Vapour density**…….0.5mark**

MM=2 X 14

MM=28

Molecular mass of a compound is 28**……..0.5mark**

Third stage: To calculate molecular formula of a compound

$M.F = n(E.F)$ but $n = \frac{MM}{mw\ of\ E.F}$……***0.5mark***

$$n = \frac{28}{12 + (1X2)}$$

$$n = \frac{28}{14}$$

$n = 2$……***0.5mark***

Then from $M.F = n(E.F)$……..***0.5mark***

M.F=2 (CH_2)

M.F=C_2H_6

∴The molecular formula of a compound is C_2H_6 ……***0.5mark***

The homologous series of a compound is alkane**……..01mark**

5. (a) Chemical equation is the representation of a chemical reaction by using symbols, formula or words……**01mark**
   (b) (i) Word equation

$$Magnesium\ sulphate + Sodium\ carbonate \rightarrow Magnesium\ carbonate + Sodium\ sulphate$$

Balanced chemical equation

$$MgSO_{4(aq)} + Na_2CO_{3(aq)} \rightarrow MgCO_{3(s)} + Na_2SO_{4(aq)}$$

Total ionic equation

$$Mg^{2+}_{(aq)} + SO^{2-}_{4(aq)} + 2Na^{+}_{(aq)} + CO^{2-}_{3(aq)} \rightarrow MgCO_{3(s)} + 2Na^{+}_{(aq)} + SO^{2-}_{4(aq)}$$

To cancel out all spectator ions

$Mg^{2+}_{(aq)} + \cancel{SO^{2-}_{4(aq)}} + 2\cancel{Na^{+}_{(aq)}} + CO^{2-}_{3(aq)} \rightarrow MgCO_{3(s)} + 2\cancel{Na^{+}_{(aq)}} + \cancel{SO^{2-}_{4(aq)}}$Net ionic equation

$Mg^{2+}_{(aq)} + CO^{2-}_{3(aq)} \rightarrow MgCO_{3(s)}$ **0.5mark from a balanced equation=02 total marks**

ii)Word equation

$$Sodium\ hydroxide + Copper\ (II) nitrate \rightarrow Copper\ (II) hydroxide + Sodium\ nitrate$$

Balanced chemical equation

$$2NaOH_{(aq)} + Cu(NO_3)_{2(aq)} \rightarrow Cu(OH)_{2(s)} + 2NaNO_{3(aq)}$$

Total ionic equation

$$2Na^{+}_{(aq)} + 2OH^{-}_{(aq)} + Cu^{2+}_{(aq)} + 2NO^{-}_{3(aq)} \rightarrow Cu(OH)_{2(s)} + 2Na^{+}_{(aq)} + 2NO^{-}_{3(aq)}$$

To cancel out all spectator ions

$\cancel{2Na^{+}_{(aq)}} + 2OH^{-}_{(aq)} + Cu^{2+}_{(aq)} + \cancel{2NO^{-}_{3(aq)}} \rightarrow Cu(OH)_{2(s)} + \cancel{2Na^{+}_{(aq)}} + \cancel{2NO^{-}_{3(aq)}}$

Net ionic equation

$2OH^{-}_{(aq)} + Cu^{2+}_{(aq)} \rightarrow Cu(OH)_{2(s)}$**0.5mark from a balanced equation=02 total marks**

iii) Word equation

$$Potassium\ hydroxide + Hydrochloric\ acid \rightarrow Potassium\ chloride + Water$$

Balanced chemical equation

$$KOH_{(aq)} + HCl_{(aq)} \rightarrow KCl_{(aq)} + H_2O_{(l)}$$

Total ionic equation

$$K^+_{(aq)} + OH^-_{(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \rightarrow K^+_{(aq)} + OH^-_{(aq)} + H_2O_{(l)}$$

To cancel out all spectator ions

$$K^+_{(aq)} + OH^-_{(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \rightarrow K^+_{(aq)} + OH^-_{(aq)} + H_2O_{(l)}$$

Net ionic equation

$OH^-_{(aq)} + H^+_{(aq)} \rightarrow H_2O_{(l)}$**0.5mark from a balanced equation=02 total marks**

iv)Word equation

$$Sodium\ hydrogencarbonate + Hydrochloric\ acid \rightarrow Sodium\ chloride + Water + Carbon\ dioxide$$

Balanced chemical equation

$$NaHCO_{3(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

Total ionic equation

$$Na^+_{(aq)} + HCO^-_{3(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

To cancel out all spectator ions

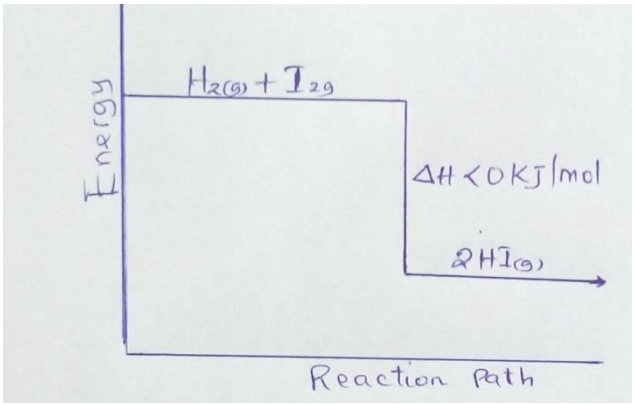
$$Na^+_{(aq)} + HCO^-_{3(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)} + H_2O_{(l)} + CO_{2(g)}$$

Net ionic equation

$HCO^-_{3(aq)} + H^+_{(aq)} \rightarrow H_2O_{(l)} + CO_{2(g)}$**0.5mark from a balanced equation=02 total marks**

6. (a)
   i. The position of equilibrium will shift from the product to the reactant because the reaction is exothermic.
   ii. No change in position of equilibrium because there is equal number of moles in the reactant and product.
   iii. The position of equilibrium will shift from the reactant to the product due to increase in concentration in the reactant side.. *02mark***@=06 total marks**

(b) ENERGY LEVEL DIAGRAM FOR $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}, \Delta H < 0KJ/mol$



…….. **03marks**

7.a) $Na_2CO_{3(aq)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$ **(02Mark)**

b) (i)

Data given

Vb = $25cm^3$

Va = $20cm^3$

Ma = 0.25M

n_a=2

n_b = 1

Mb =?

From

$\frac{M_A V_A}{M_B V_B} = \frac{n_A}{n_B}$ **……..(0½Mark)**

Mb = $\frac{MaVanb}{Vbna}$ ……(**0½Mark)**

= $\frac{0.25M \times 20cm^3 \times 1}{25cm^3 \times 2}$

= 0.1M

The Molarity of anhydrous sodium carbonates 0.1M…..**(01Mark)**

ii) To calculate concentration of a base

Mass=7.15g

Volume=$250cm^3 = 0.25dm^3$**…..0.5mark**

C=?

$Concentration = \frac{mass}{volume}$**……..0.5mark**

$conc = 7.15/0.25$

Conc=28.6g/dm^3 **……..01 mark**

To calculate molar mass of a base

Molarity=concentration/ molar mass**….0.5mark**

Molar mass=concentration/molarity

Molar mass=28.6/0.1

Molar mass =286g/mol…….**01mark**

To calculate the value of X

$Na_2CO_3.XH_2O$=Molar mass……**0.5mark**

$Na_2CO_3.XH_2O$=286

(23x2)+ (12x1)+(16x3) + X((1x2) + (16x1) =286

46+12=48+ X(2+16)=286

106+18X=286

106-106+18X = 286-106

18X=180

18 18

X = 10

∴ The value of X = 10 ……**(01Mark)**

8. (a)

i. The concentration of sulphuric acid will be lowered because the concentrated sulphuric acid will undergo hygroscopic process.

ii. Reddish brown fumes of nitrogen dioxide will be observed because silver nitrate will be decomposed since it is unstable to light.

$$2AgNO_{3(s)} \xrightarrow[\Delta]{} 2Ag_{(s)} + 2NO_{2(g)} + O_{2(g)}$$

$$Reddish\ brown\ fumes$$

iii. White fumes of ammonium chloride will be observed.

$$HCl_{(g)} + NH_{3(g)} \rightarrow NH_4Cl_{(g)}$$

$White\ fumes$ *1mark***@=03 total marks**

(b)

i. Due to the formation of protective layer which is lead (II) chloride that prevents further reaction.

$$Pb(NO_3)_{2(aq)} + 2HCl_{(aq)} \rightarrow PbCl_{2(s)} + 2HNO_{3(aq)}$$

ii. Zinc reacts with dilute hydrochloric acid because it Is a more reactive element compared to hydrogen in the reactivity series but copper does not react with dilute hydrochloric acid because it is less reactive than hydrogen in the reactivity series.

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$$

$$Cu_{(s)} + HCl_{(aq)} \rightarrow No\ reaction$$

iii. The reaction is exothermic but also magnesium is more reactive than carbon in the reactivity series than carbon.

$2Mg_{(s)} + CO_{2(g)} \rightarrow 2MgO_{(s)} + C_{(s)}$.

*1mark for explanation and 01mark for balanced chemical equation=**06 total marks***

SECTION C

9. a)i)when water has colour **1mark**

ii)when water it smells **1mark**

b)Introduction**1.5 marks**

points(warning to the likely effects)

-it accelerate to deforestation

-global warming

-eruption of diseases caused by the product of coal combustion **10marks@2**

-disappearance of valuable plant species

-it can lead to creation of pits in the ground

Conclusion **1.5marks**

10. Ore used-Haematite……**0.5mark**

Method used during extraction-Chemical reduction…….**0.5mark**

Materials fed in the blast furnace:Haematite, coke, and limestone……..**0.5mark**

Functions of haematite, coke and limestone

Haematite-acts as a chief ore of iron:……..0.5mark

Coke-acts as a reducing agent by reducing carbon dioxide into carbon monoxide**:……..0.5mark**

Limestone-used to remove acidic impurities:……..**0.5mark**

Chemical equations taking place in the blast furnace

i. Formation of carbon dioxide

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$……**01mark**

ii. Formation of carbon monoxide

$CO_{2(g)} + C_{(s)} \rightarrow 2CO_{(g)}$……**01mark**

iii. Reduction of haematite

$Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(s)} + 3CO_{2(g)}$……**01mark**

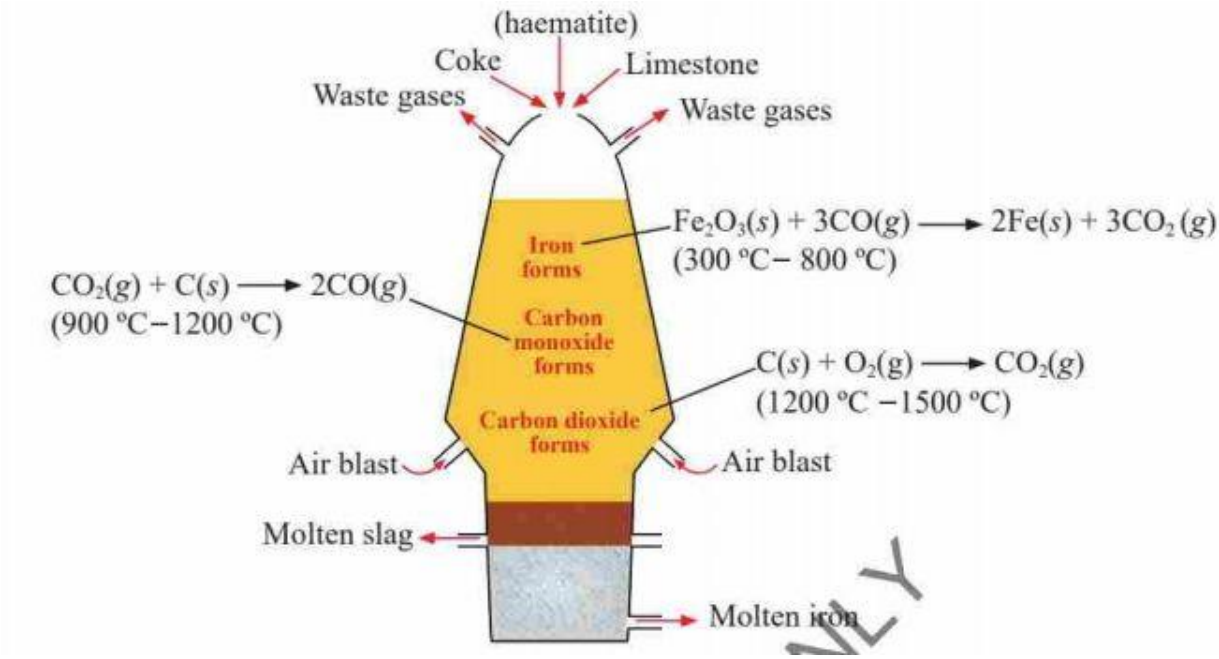
iv. Decomposition of limestone

$CaCO_{3(s)} \xrightarrow[\Delta]{} CaO_{(s)} + CO_{2(g)}$……**01mark**

v. Formation of slag

$CaO_{(s)} + SiO_{2(s)} \rightarrow CaSiO_{3(l)}$……**01mark**

Diagram



Drawing=**03marks,** labeling at least six parts=**03marks**

At the bottom, there are two liquids produced which are; molten iron ad slag but slag is removed as the impurities……**01mark.**

11. a) From the reaction

$$2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$$

For sulphur dioxide (SO_2)

$2mol\ of SO_2 = 2mol\ of\ SO_3$……**0.5mark**

$$3mol\ of\ SO_2 = X$$

$$X = \frac{3mol\ of\ SO_2\ X2mol\ of\ SO_3}{2mol\ of SO_2}$$

$X = 3mol\ of SO_3$……..**01mark**

For oxygen gas (O_2)

$1mol\ of O_2 = 2mol\ of\ SO_3$……**0.5mark**

$$4mol\ of\ O_2 = Y$$

$$Y = \frac{4mol\ of\ O_2\ X2mol\ of\ SO_3}{1mol\ of O_2}$$

$Y = 8mol\ of O_2$……**01mark**

$\therefore$ *Sulphur dioxide is present in smaller amount because it produce less amount of sulphur trioxide.*

01 mark

b) Reactant left in oxygen gas……**0.5mark**

$1mol\ of O_2 = 2mol\ of\ SO_3$……**0.5mark**

$$X = 3mol\ of\ SO_3$$

$$X = \frac{3mol\ of\ SO_3 X 1mol\ of\ O_2}{2mol\ of SO_3}$$

$$X = 1.5mol\ of O_2$$

$Reactant\ left\ in the\ container\ have\ 1.5mol$……..**01mark**

To calculate amount of reactant remained

Amount remained=Original amount-Amount remained……**01mark**

Amount remained=4mol-1.5mol

∴Amount remained=2.5mol

Amount of reactant remained in the container is 2.5mol……..**01mark**

To calculate mass

Data

Number of moles(n)=2.5mol

Molar mass (mr) of O_2=(16X2)

=32g/mol……**0.5mark**

Required: mass(m)

$From\ the\ formula: Number\ of\ moles\ (n) = \frac{mass(m)}{molar\ mass(mr)}$…….**01mark**

$$2.5 = \frac{m}{32}$$

$$m = 2.5\ X\ 32$$

$$m = 80g$$

$\therefore Mass\ of\ reactant\ left\ in\ the\ reactant\ is\ 80g$…….**01mark**

c) Number of sulphur trioxide is obtained from limiting reactant which is sulphur dioxide……**0.5mark**

$$2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$$

$For\ sulphur\ dioxide\ (SO_2)$

$2mol\ of SO_2 = 2mol\ of\ SO_3$

$3mol\ of\ SO_2 = X$ ……..**0.5mark**

$$X = \frac{3mol\ of\ SO_2\ X 2mol\ of\ SO_3}{2mol\ of SO_2}$$

$$X = 3mol\ of SO_3$$

$\therefore There\ are\ 3moles\ of\ sulphur\ trioxide\ produced$……..**01mark**

$d) Data$

Number of moles(n)= $3mol\ of SO_3$

GMV at S.T.P=$22.4dm^3$…….**0.5mark**

Required: Volume (V)

$From\ the\ formula: Number\ of\ moles\ (n) = \frac{Volume\ (V)}{GMV\ at\ S.T.P}$……..**01mark**

$$3 = \frac{V}{22.4}$$

$$V = 3X22.4$$

$$V = 67.2dm^3$$

$\therefore There\ are\ 67.2\ litres\ of\ sulphur\ trioxide\ produced\ at\ STP$…….**01mark**